



SIGNAL CARTEL

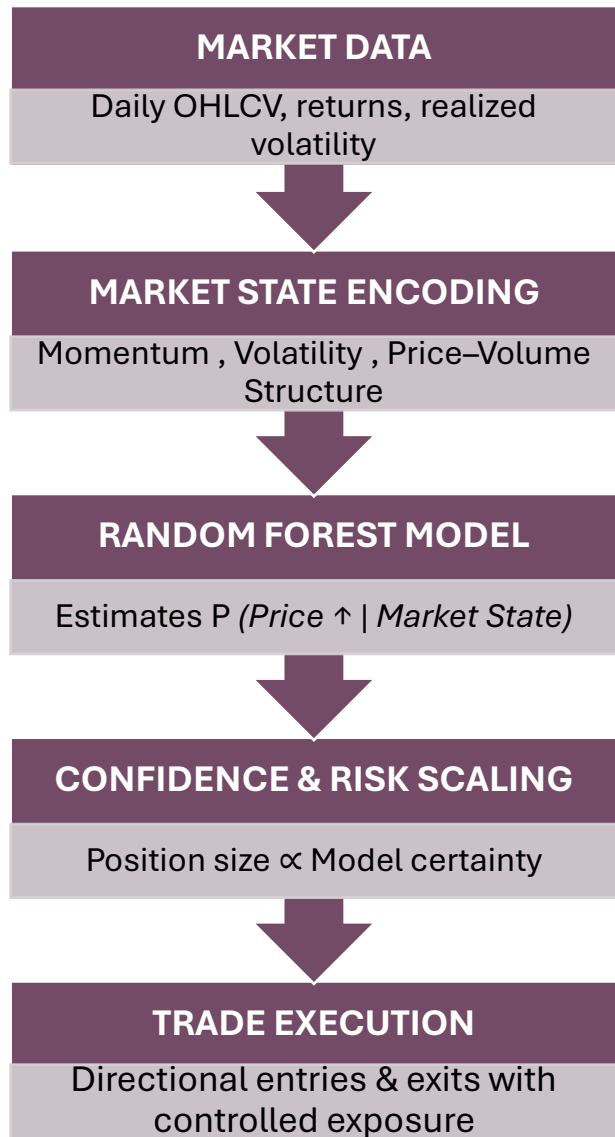
A PROBABILISTIC DIRECTIONAL TRADING STRATEGY

A Random Forest based direction modelling
using price-volume structure and volatility-
adjusted position sizing

STOCK : SONATSOFTW.NS

*from constrained Universe





STRATEGY INTUITION: PROBABILISTIC DIRECTIONAL EDGE

- **CORE HYPOTHESIS**

Short-horizon markets exhibit **weak but persistent** directional biases driven by price-volume dynamics and volatility regimes.

- **WHY RANDOM FOREST?**

Financial Signals are non-linear, noisy, and regime-dependent, requiring a model that prioritizes robustness over point accuracy.

- **WELL SUITED BECAUSE :**

- Captures **non-linear interactions** without assuming functional form
- Is resilient to noise and feature collinearity
- It Aggregates multiple weak predictors into a stable probability estimate
- This makes it well-suited for **directional probability estimation, not point forecasting.**

TRADING LOGIC: FROM PROBABILITY TO POSITION

SIGNAL DEFINITION:

The strategy trades on the model-estimated directional probability:

$$P(\textit{Direction} \mid \textit{Market State})$$

TRADE DIRECTION

- **Long** if upward probability > entry threshold
- **Short** if downward probability > entry threshold

ENTRY CONDITIONS

Enter a trade only when:

- Probability exceeds a predefined confidence threshold
- Signal aligns with current market state

Effect:

Selective participation in high-signal regimes

EXIT CONDITIONS

Exit a position when:

- Model confidence drops below exit threshold
- Signal direction reverses
- Maximum holding horizon is reached

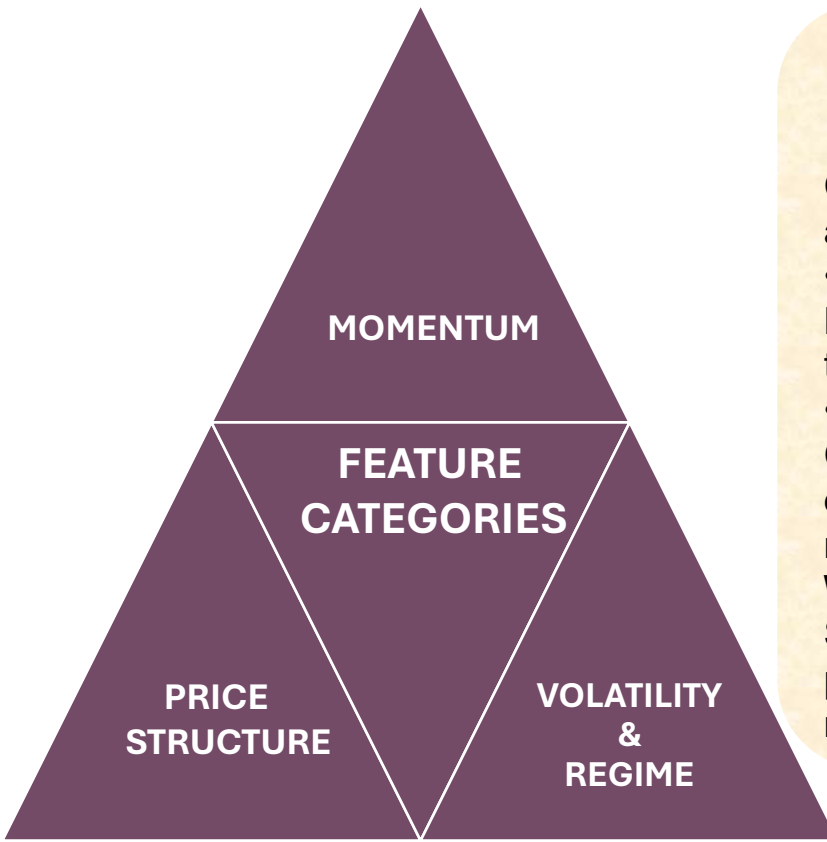
Effect:

Prevents overstaying decayed signals and regime transitions

FEATURE DESIGN: ENCODING MARKET STATE

The model does not rely on raw prices.

Instead, it learns from **transformed features that capture momentum, volatility, and price-volume structure.**



➤ **MOMENTUM FEATURES:**

Captures trend persistence and direction strength.

- **MACD Histogram**

Measures short-term vs long-term momentum imbalance

- **Lagged Returns**

Captures immediate directional continuation or reversal

Why momentum?

Short-horizon trends often persist due to delayed market reaction.

➤ **VOLATILITY AND REGIME FEATURES:**

Captures market uncertainty and regime shifts.

- **Rolling Volatility (20-day)**

Measures risk regime (calm vs turbulent)

- **High-Low Range / Close**

Captures intraday uncertainty and price compression/expansion

Why volatility?

Directional signals behave differently in low vs high volatility regimes.

➤ **PRICE STRUCTURE FEATURES:**

Captures relative price positioning.

- **RSI (14)**

Identifies overbought / oversold pressure

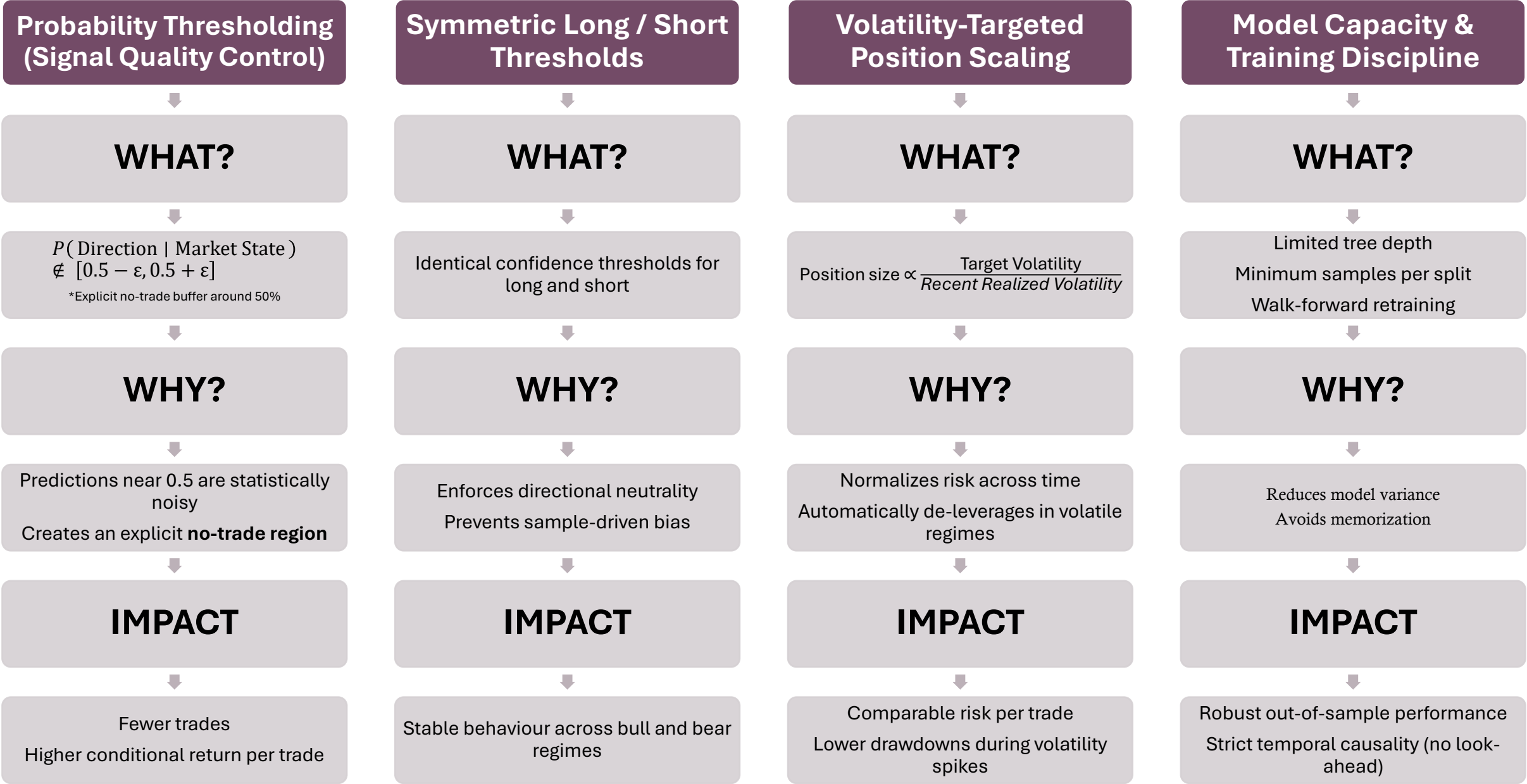
- **Bollinger %B**

Normalized price position within volatility bands

Why structure?

Directional edges depend on *where* price is, not just where it's going.

THRESHOLDS & PARAMETER DESIGN LOGIC



RISK MANAGEMENT & DRAWDOWN CONTROL

Risk is controlled structurally via signal selectivity, volatility scaling, and disciplined exits. Not reactive stop-losses.



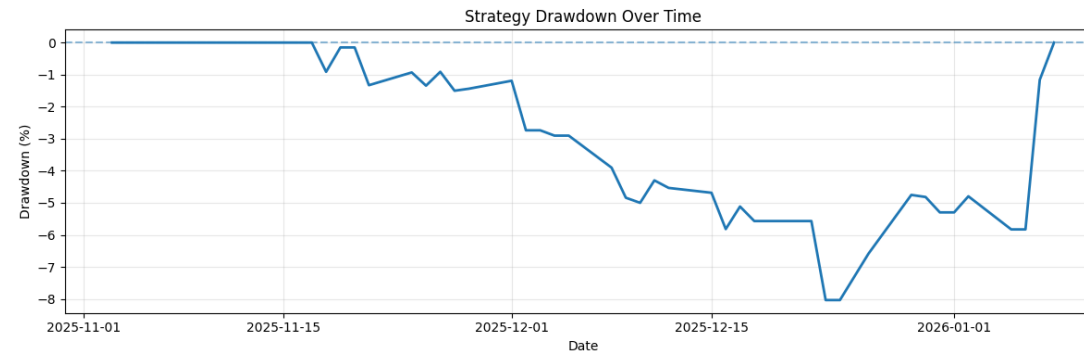
Explicit no-trade region around uncertainty



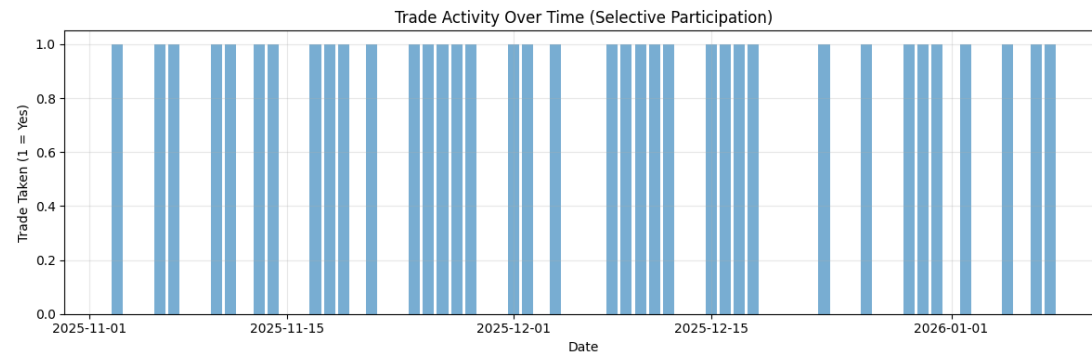
Risk normalized across volatility regimes



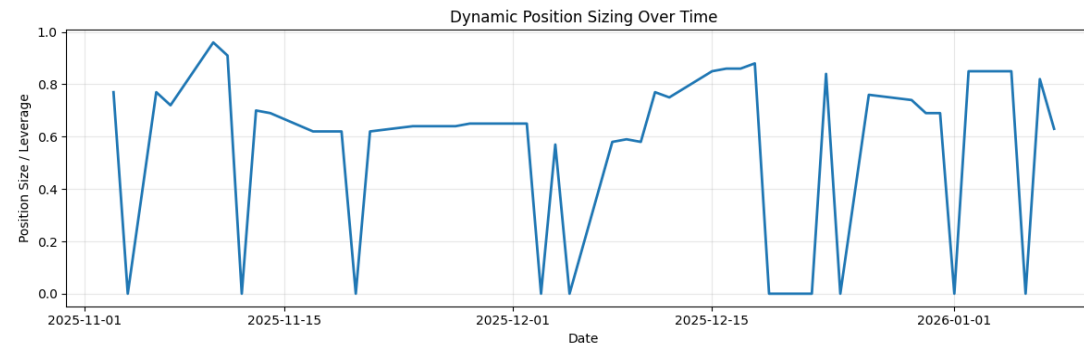
Hard upper bound on exposure



- **Drawdowns are structurally bounded**
De-leveraging during volatility spikes prevents drawdown acceleration.

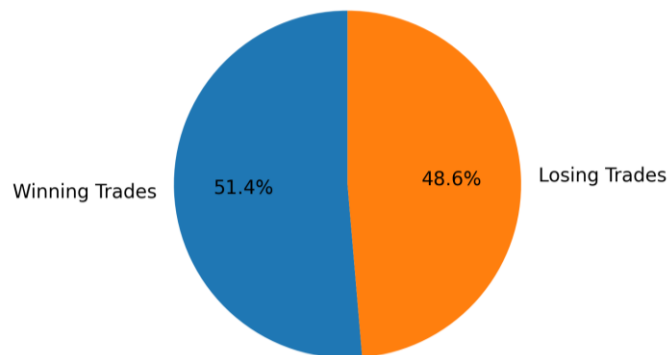


- **Selective participation**
Trades occur only when model confidence exceeds threshold, creating a natural no-trade zone.



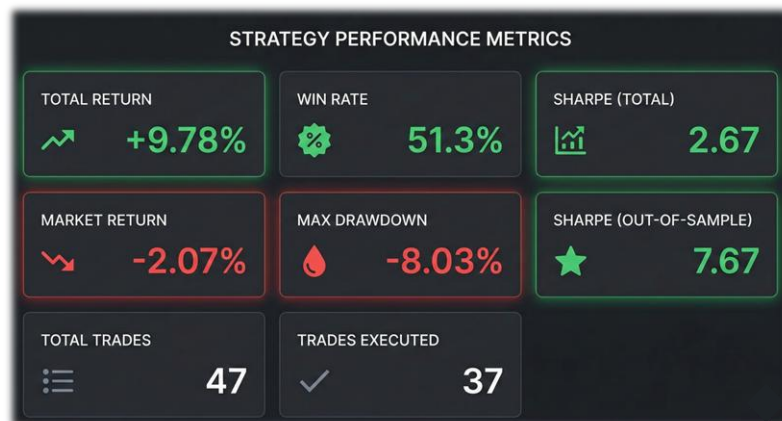
- **Automatic Risk Scaling**
Position size contracts during high volatility and expands in stable regimes.
*Volatility-targeted exposure

RESULTS & TRADE QUALITY

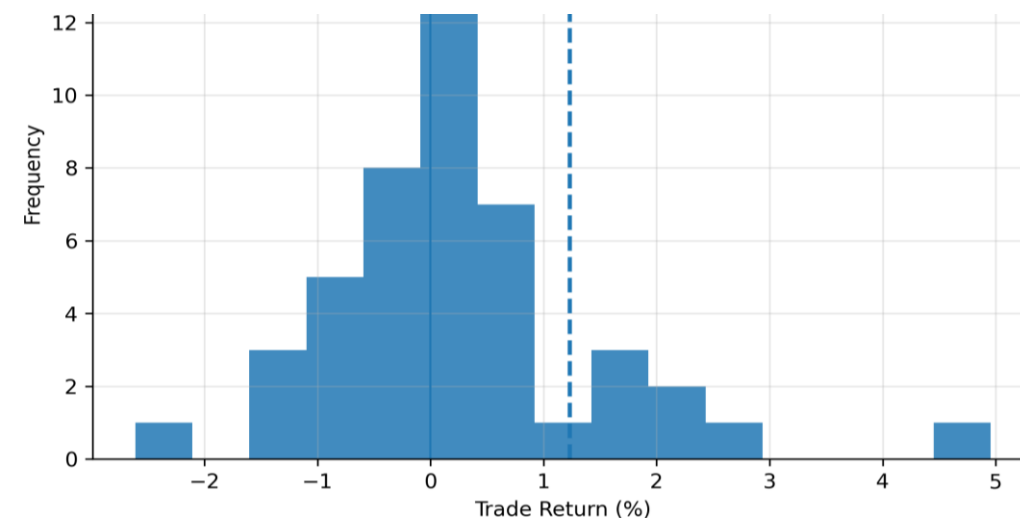


Win vs Loss Breakdown

- Win rate is healthy but **not the edge**
- Strategy focuses on **expectancy**, not hit-rate optimization

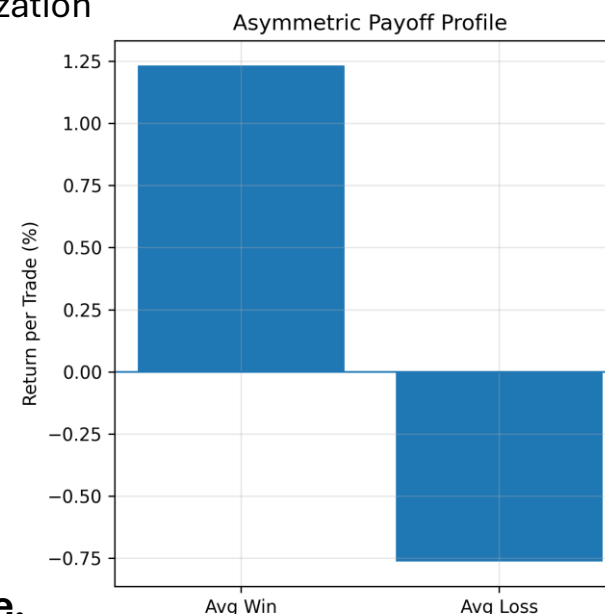


Outperformance is structurally driven by payoff asymmetry and selective participation, not by high win rate or aggressive leverage.



Per-Trade P&L Distribution:

- Centered near zero with **right-skewed payoff distribution**
- Indicates controlled losses and occasional outsized gains
- Positive mean P&L confirms persistent edge



Average Win vs Average Loss

- Average win > Average loss
- **Positive trade-level expectancy observed**
Losses tightly controlled, gains occasionally outsized
- **Returns not dependent on high win rate**
Profitability driven by payoff size, not frequency.
- **Right-skewed payoff distribution**
Larger winners dominate cumulative returns.

1

TEMPORAL INTEGRITY

Walk-forward training ensures every prediction uses only past data.
No look-ahead, no leakage, deployment-faithful by construction.

2

Out-of-Sample STRENGTH

Sharpe improves out-of-sample, indicating the signal is not overfit
and stabilizes under unseen market conditions.

3

CAPACITY CONTROL

Random Forest depth capped and minimum samples enforced to prevent
memorization of short-term noise.

4

REGIME ROBUSTNESS

Strategy adapts exposure across volatility regimes instead of
forcing constant market exposure.

RISK & PERFORM ANCE COMPARI SON

IN -
SAMPLE
(IS)

OUT-OF-
SAMPLE
(OS)

Sharpe
Ratio

1.33

7.67

Max
Drawdown

-8.03%

-1.08%

- Drawdown *contracts* out-of-sample
- Performance improves **without increased tail risk exposure**
- Confirms structural generalization, not curve-fit behavior

ROBUSTNESS & GENERALIZATION

Design choices that prevent overfitting

LIMITATIONS & SCOPE OF INFERENCE

Contextual constraints imposed by the problem statement

LIMITATIONS

1

Short Backtest Horizon (Problem-Imposed)

- Data window restricted to ~2 months (1 Nov 2025 – 31 Dec 2025)
- Limits exposure to **few market regimes**
- Restricts long-horizon statistics (e.g., crisis behaviour)
- Backtest length is **not a modeling choice**

2

47 total days, 37 executed trades

- Performance metrics have **higher variance**
- Sharpe and drawdown estimates are **directional, not asymptotic**
- Emphasis shifts from magnitude → **behavioral consistency**

3

Strategy tested on **one NSE-listed equity**

- No cross-sectional diversification
- Results reflect **signal quality**, not portfolio construction
- Generalization tested **temporally**, not cross-sectionally

FUTURE SCOPE & ROADMAP Moving from Signal Generation to Fund Management

1

From Single-Stock to Portfolio:

- Deploy strategy across Nifty 50 universe to diversify idiosyncratic risk.

2

Dynamic Regime Adaptation:

- Instead of fixed thresholds (0.55),
- implement Adaptive Thresholding based on real-time market volatility

3

Production Hardening:

- Deploy Slippage Modelling and Transaction Cost Optimization to simulate realistic institutional trading environments.

STRATEGY DNA & FINAL TAKEAWAY

Probabilistic • Selective • Risk-Normalized

- This strategy does **not** predict prices.
- It estimates when odds are favorable, participates selectively, and scales risk **ex-ante**.

Core Design Principles

- **Probabilistic Validation**
Predictions are acted upon only when confidence exceeds noise, creating a natural no-trade region.
- **Structural Risk Control**
Risk is normalized through volatility-targeted sizing and hard leverage caps — not reactive stop-losses.
- **Deployment-Faithful Modeling**
Walk-forward training ensures strict temporal causality and eliminates look-ahead bias by construction.

WHAT RESULTS ACTUALLY PROVE

- Performance is driven by **payoff asymmetry**, not **win-rate inflation**.
- Drawdowns **contract out-of-sample**, confirming robustness
- Risk does **not increase** with performance
- Signal strength improves under **unseen market conditions**

Designed for execution. Built for uncertainty.